

Product Requirement Document for Medicine Dispensing Device

First-Generation Removable-Compartment Product

Field	Value
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Status	Draft
Version	1.4
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Mechanical basis	PrototypeDesignV2 - removable individual compartments with magnetic retention and a fixed hand-off/removal station
Revision purpose	Replace the prior gravity-drop/chute concept with the new removable-compartment dispensing architecture and align software workflows accordingly.

Revision History

Version	Change	Summary
1.0	Initial PRD	Original product scope.
1.1	Architecture revision	Clarified software responsibilities and UI-first build strategy.
1.2	Capacity revision	Established first-generation capacity target.
1.3	Integrated hardware/software revision	Combined prior rotary-carousel, gravity-dispense hardware with software architecture.
1.4	Mechanical Design V2	Adopts 11 removable individual dose compartments on a rotating carrier with magnetic retention; dispensing is by presenting/removing the compartment rather than dropping pills through a chute.

1. Overview

1.1. Purpose

This document defines the product requirements for a personal, automated desktop medication dispensing device. The first-generation product uses removable individual dose compartments mounted on a rotating carousel. A scheduled compartment is first visually inspected, then moved to a fixed hand-off/disbursement opening where the patient removes the entire compartment. The device subsequently verifies whether the compartment was removed and records the outcome.

- a. Store scheduled medication in up to 11 removable individual dose compartments.
- b. Allow a patient or caregiver to define medication schedules and load/refill each removable compartment.
- c. Automatically position the correct compartment at the visual-inspection station when a dose is due.
- d. Visually verify the expected medication contents before the compartment is made available to the patient.
- e. After successful verification, move the same compartment to the hand-off/disbursement station for removal by the patient.
- f. After a configured pickup interval, determine whether the compartment was removed and alert the patient/caregiver when it remains present.
- g. Provide local and remote-capable interfaces for configuration, monitoring, alerts, loading/refill and adherence history.

1.2. First-Generation Capacity and Mechanical Configuration

The Mechanical Design V2 uses 11 removable wedge-shaped medication compartments retained on a rotating carrier by magnets mounted to the inner wall, with a corresponding metal feature on each removable compartment. One angular station is reserved as the fixed hand-off/removal opening. The resulting first-generation medication capacity is 11 scheduled dose compartments at one time.

Capacity in days depends on dose frequency. At three doses per day, 11 compartments provide three complete days plus two additional scheduled doses. The product shall therefore describe capacity primarily in number of dose compartments, not as a fixed number of days.

1.3. Problem Statement

The device is intended to reduce missed doses and incorrect medication use while lowering caregiver burden. The removable-compartment design additionally provides the patient with a self-contained dose container rather than loose pills, simplifying patient pickup and eliminating the need for a gravity chute/drop mechanism.

1.4. Product Design Principles

- Verify before hand-off: a compartment is not presented to the patient until its contents pass the configured inspection criteria or an authorized override is recorded.
- Present the complete dose container: medication remains inside its removable compartment during patient hand-off.
- Deterministic positioning: the device must establish a known carousel position before automatic operation.
- Positive compartment state: the system tracks whether each removable compartment is installed, loaded, presented or removed.
- Controlled patient access: normal patient interaction occurs only at the fixed hand-off/removal opening.
- Application-layer orchestration: scheduling, vision, hardware control, persistence and notifications remain separate subsystems coordinated by workflows.

2. Actors and User Experiences

Actor	Primary interactions
Patient	Receives a dose by removing the presented individual compartment; acknowledges pickup if requested; receives local alerts.
Caregiver	Defines medication/schedules, loads and reinstalls removable compartments, reviews inspection/pickup history, resolves alerts and may monitor remotely.
Service/Developer	Tests motion, camera, sensors, compartment presence and firmware/software interfaces; not a normal end-user role.

3. Physical Product Concept

3.1. Rotating Carrier and Removable Compartments

- a. The carrier shall provide indexed positions for 11 removable dose compartments plus one fixed access/hand-off station.
- b. Each removable compartment shall have a unique human-readable identifier corresponding to its logical compartment ID.
- c. Each removable compartment shall include a ferromagnetic/metal retention feature that mates with a magnet fixed to the rotating carrier's inner wall.
- d. Magnetic retention shall be strong enough to prevent unintended compartment movement during normal rotation but weak enough for intentional manual removal at the hand-off station.
- e. The rotating assembly shall include the compartment-supporting base and separating walls needed to locate each removable compartment.
- f. The fixed device structure shall provide the camera/inspection geometry and the hand-off access opening.

3.2. Inspection Station

- a. A fixed camera shall capture the medication contents of the target removable compartment before hand-off.
- b. Lighting and camera geometry shall be controlled to support repeatable visual verification.
- c. The inspection station shall be positioned so that the removable compartment remains fully retained and cannot be accidentally removed during inspection.

3.3. Hand-Off / Disbursement Station

- a. After successful visual verification, the same removable compartment shall be rotated to a fixed patient-accessible hand-off opening.
- b. The patient shall remove the complete compartment rather than receiving loose medication through a chute.
- c. The system shall prevent carousel motion while a compartment is partially removed or while the hand-off condition is otherwise unsafe.
- d. The device shall determine whether the presented compartment remains installed after the configured pickup interval, using camera-based inspection and/or a dedicated presence sensor.

4. Functional Requirements

4.1. Medication and Schedule Management

- a. Schedule shall support medication, quantity, time-of-day, recurrence/day rules, start date and end date where applicable.
- b. System shall support one-time and recurring dose patterns and timezone-aware evaluation.
- c. System shall prevent invalid duplicated/overlapping schedules.
- d. Schedule changes shall not silently change the physical contents or identity of already-loaded removable compartments.

4.2. Loading and Refill

- a. Caregiver shall be guided to load a specific identified removable compartment with the scheduled medication(s).
- b. The UI shall show compartment ID, expected medication(s) and quantity during loading.
- c. The caregiver shall reinstall the compartment into its intended carrier position and the system shall verify that the compartment is present before it is considered ready.
- d. The system should support visual verification of loaded contents before normal scheduled operation.
- e. Removed/used compartments shall be clearly represented as unavailable until returned, reloaded and verified.

4.3. Scheduled Dose Presentation

- 1. Scheduler identifies a due dose and its assigned removable compartment.
- 2. Application verifies device readiness and known carousel position.
- 3. Carrier rotates the assigned compartment to the inspection station.
- 4. Camera captures an image and the Vision Service verifies expected contents.
- 5. If verification fails, the dose is blocked and the compartment is not presented for patient removal.
- 6. If verification passes, the carrier rotates the same compartment to the hand-off/disbursement opening.
- 7. Patient is notified that the dose compartment is ready for pickup.
- 8. After the configured pickup interval, the device determines whether the compartment has been removed.
- 9. If removed, the dose is recorded as COLLECTED and the physical compartment is marked REMOVED/UNAVAILABLE until reloaded.
- 10. If still present, the system records NOT_COLLECTED/MISSED according to configured timing and notifies the appropriate user.

4.4. Image-Based Medication Verification

- a. The system shall capture an image before each scheduled compartment is presented for removal.
- b. Vision shall compare the captured contents with the expected medication using available visual features such as shape, color, imprint, size and count.
- c. Low-confidence or mismatched results shall block automatic hand-off unless an authorized override is recorded.
- d. Vision shall also support presence/absence determination of a presented compartment when used for pickup verification.

4.5. Compartment Presence and Pickup

- a. The system shall maintain a presence state for each removable compartment.
- b. A presented compartment shall remain logically associated with its scheduled dose until pickup is verified or the pickup window expires.
- c. The device shall not interpret a compartment as collected merely because it was moved to the hand-off station.
- d. If camera-based pickup verification is inconclusive, the device shall enter an exception state and request user/caregiver confirmation rather than assuming pickup.

4.6. Safety and Interlocks

- a. Automatic carousel motion shall be inhibited whenever the system cannot establish a safe hand-off state.
- b. The product-intent design shall protect the patient from contact with moving carousel components except at the intended removable-compartment access opening.
- c. The system shall detect or conservatively handle a partially removed/misaligned compartment before rotation.
- d. On power loss or position uncertainty, the device shall re-establish carousel reference before automatic operation resumes.

5. Software Product Requirements

5.1. Application Architecture

The product shall use a modular Python application with an Application/Workflow layer coordinating scheduling, inventory/compartment state, vision, hardware control, logging and notification. The scheduler determines which dose is due but does not directly command the carousel.

5.2. Core Product Data

Entity	Purpose
Patient	Patient identity, timezone and active state.
Medication	Canonical medication identity and descriptive information.
Schedule / ScheduledDose	Dose timing and concrete due-dose occurrences.
RemovableCompartment	Physical compartment identity, carrier position, presence and lifecycle state.
CompartmentMedication	Expected/loaded medication and quantity.
InspectionEvent	Pre-hand-off medication verification or post-hand-off presence verification.
PresentationEvent	When a verified compartment is positioned at the hand-off station.
PickupEvent	Verified removal, timeout/not-collected result or manual resolution.

Entity	Purpose
Alert / SystemEvent	Auditable alerts, overrides, hardware faults and workflow history.

5.3. User Interface

- a. Dashboard shall show next dose, current device state, active alerts and number of installed/available compartments.
- b. Carousel view shall show all 11 compartment IDs and states, plus the fixed inspection/hand-off station status.
- c. Compartment detail shall show expected contents, current presence, latest pre-dose inspection and pickup history.
- d. Guided refill shall show which removed compartments need to be returned/reloaded.
- e. Development mode shall support simulated compartment removal and fake hardware adapters.

6. Notifications and Audit History

- a. Alerts shall include medication-verification failure, compartment missing before due time, compartment not removed within the pickup window, partial/misaligned compartment condition, homing/position fault, and other device failures.
- b. Event history shall record schedule changes, compartment loading/reinstallation, pre-dose inspection, hand-off presentation, pickup verification, manual overrides and hardware faults.
- c. Manual override shall be explicit, authorized and auditable.

7. First Integrated Prototype Acceptance Criteria

Area	Acceptance objective
Compartment retention	All 11 installed compartments remain seated through repeated normal carousel movements.
Manual removal	A patient can intentionally remove the presented compartment from the hand-off station without tools and without disturbing adjacent compartments.
Indexing repeatability	Every compartment can repeatedly reach inspection and hand-off positions without cumulative drift.
Inspection repeatability	Medication is consistently framed under controlled camera/lighting conditions.
Pickup detection	System reliably distinguishes compartment present versus removed at the hand-off station.
Partial-removal safety	Carousel does not move when the presented compartment is in an unsafe intermediate position.
Recovery	After power interruption, the device establishes a known reference and reconciles installed/missing compartments.

Area	Acceptance objective
End-to-end workflow	Due dose -> inspect -> verify -> present -> pickup verification -> event/UI update completes deterministically.

8. Key Changes from PRD v1.3

Previous design	Mechanical Design V2
12 dose compartments	11 removable dose compartments plus one fixed access/hand-off station.
Medication released through a bottom aperture	Medication remains inside the removable compartment.
Gravity chute and receiving bucket	Patient removes the complete compartment.
Drop sensor confirms dispensing	Compartment-presence/removal verification confirms pickup.
DISPENSED / TAKEN flow	PRESENTED / COLLECTED flow.
Stationary floor/aperture is critical	Magnetic retention, removable-compartment alignment and access interlock become critical.